SKALD: User Manual & Technical Guide

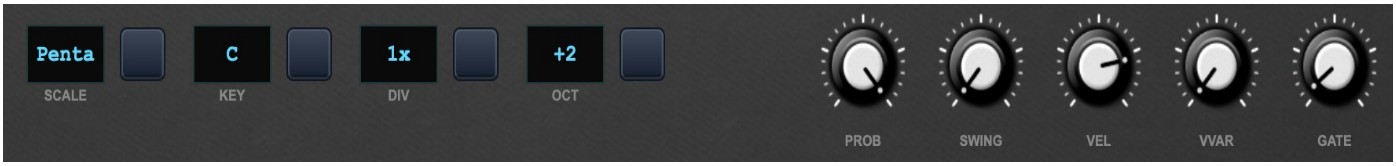
1. Introduction & Core Concept

Skald is a generative "Turntable" MIDI sequencer inspired by the organic movement of physical objects and the mathematical beauty of Euclidean rhythms.

Unlike linear step-sequencers, Skald operates on a circular logic. Imagine a vinyl record where notes are "dots" placed on concentric tracks. As the record rotates, these dots pass under a stationary "sensor arm" at the top (12 o'clock position), triggering MIDI notes.

The goal of Skald is to provide a performance-oriented environment where rhythm is discovered through geometry, probability, and rotation, bridging the gap between mechanical precision and human-like variation.

2. Interface Elements & Functions

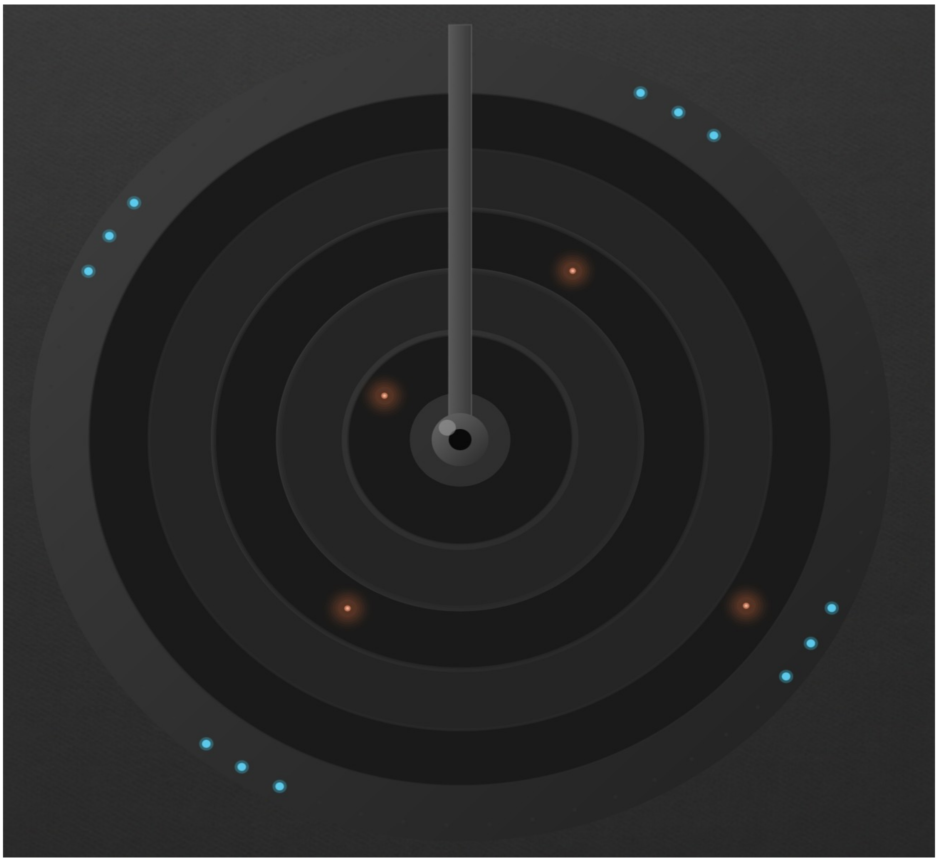


Top Control Bar

- **LED Displays (Scale, Key, Speed, Octave)**: These high-visibility displays show your current harmonic and temporal settings. Click the **TAP** buttons next to them to cycle through scales or change quantization divisions.

- **Performance Knobs**:
  - **Probability**: Determines the chance (0-100%) that a note will actually trigger when passing the arm. Great for adding "breathing" to a pattern.
  - **Swing**: Introduces micro-timing offsets to "off-beat" notes for a human feel.
  - **Velocity & Variation**: Sets the base hit strength and adds a random range to prevent robotic repetition.
  - **Gate Time**: Controls the duration of the MIDI note-on message (from short staccatos to long drones).

The Turntable (Main Area)



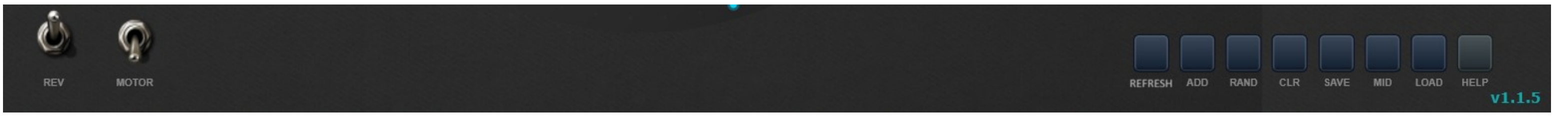
- **The Rings**: There are 10 zones (1 Scratch Ring, 8 Note Rings, 1 Center).
- **Visual Feedback**: Dots glow when passing the arm. A "Tracer" effect shows the active Gate Time, fading out as the note-off signal is sent.

Bottom Action Buttons

- **REFRESH**: Scans your system for MIDI output ports (Essential in Standalone mode).
- **ADD**: Enters "Add Mode" to place new dots on the rings.
- **RAND**: Generates a random scattering of dots for instant inspiration.
- **CLR (CLEAR)**: Wipes the current pattern.
- **SAVE/LOAD**: Manages your pattern presets (.ttp files).

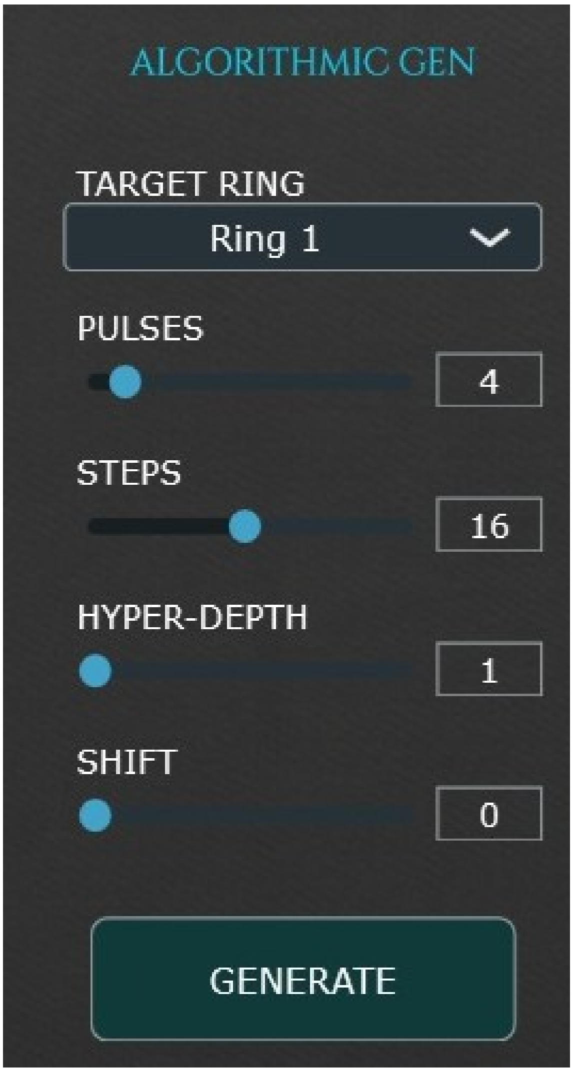
  SAVE to save the pattern configuration, including the dots, and **LOAD** to load it into the app. You can choose any directory. Presets have a .ttp extension.
- **MID (Export)**: Clicking this button allows you to export the current circular pattern as a standard MIDI file (.mid). A file browser will open, allowing you to **save the file to any location on your hard drive**. Once saved, you can manually import or drag this file into your DAW's timeline to further edit the performance or use it with other virtual instruments.
- **HELP:** this open a pdf manual (place the user guide in the same folder as standalone or plugin and not change his name (Skald_Manual.pdf).

The higher and the lower areas of the app in the standalone version 1.1.5 seen in its full horizontal extension looks like this:



Algorithmic Gen Sidebar (Hyper-Euclidean)

The Algorithmic Gen section allows you to create complex patterns based on Euclidean geometry. This section allows you to generate mathematical patterns (Euclidean or based on density algorithms) in just a few seconds.



1. **Ring Selector**:
   Choose which ring (note) to operate on.
   Select which ring to apply the generation to.
2. **Pulses (*)**:
   Set as many notes or strokes as you want.
3. **Steps (*):**
   Sets the total number of steps over which to distribute the notes.

   (*) Pulses & Steps: They define how many strokes to distribute evenly in a number of subdivisions (e.g. 3 strokes over 8 steps). Generates mathematically perfect Euclidean rhythms on the selected ring
4. **Hyper-Depth**:
   Distributes notes across adjacent rings for melodic complexity. Adjusts pattern complexity. A special Skald feature that layers patterns for greater rhythmic complexity.
5. **Shift:**
   Adjusts the rotational offset of the pattern.
   Rotates the generated pattern to move the accents relative to the start of the cycle.
6. **Button "Generate"**:
   Applies the Hypereuclidean algorithm to the turntable to generate hypereuclidean patterns. Applica istantaneamente il pattern calcolato all'anello selezionato.

In Norse folk belief and culture, skalds (Old Norse: skald) were court poets active between the 9th and 11th centuries, figures of the highest prestige who composed and recited verses for kings, chieftains, and nobles. They were considered the guardians of the historical, cultural, and spiritual memory of the Scandinavian peoples.

3. Note Properties & MIDI Assignment

Skald allows deep per-note customization using **Modifier Keys** while dragging a dot:

- **Placement**: Click **ADD**, then click any ring to place a dot. Drag it to change its timing (angle).
- **MIDI Channel (ALT + Drag)**: Vertical movement changes the MIDI channel (1-16). The channel number is displayed directly on the dot. This allows one instance of Skald to trigger multiple instruments simultaneously.
- **Velocity (SHIFT + Drag)**: Adjusts the individual volume/intensity of that specific dot.
- **Gate Time (CTRL/CMD + Drag)**: Adjusts how long that specific note stays "pressed."
- **Selection**: Click a dot to select it; the HUD (Heads-Up Display) in the top right will show precise values for its parameters.

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4. Standalone vs. Plugin Mode

- **Standalone**: Features a dedicated **Transport Bar** (Play/Stop and BPM Slider) at the top. Use the **REFRESH** button to select your physical MIDI interface or Virtual MIDI Cable (like loopMIDI or IAC). You can choose the midi outputs from the options (button at the top left) or do a "refresh" of the midi drivers of your system through the dedicated button at the bottom right and from here choose your output port to which the channels of your dots will be directed.
- **Plugin (VST3/AU)**: The transport is synced to your DAW. The BPM and Play/Stop functions are hidden as Skald follows the host's clock perfectly. MIDI is sent directly to the track's output or to other tracks via DAW routing.

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5. DAW Setup Guide

To use Skald as a MIDI effect, follow these routing steps:

- **Ableton Live**: Place Skald on a MIDI track. Create a second MIDI track with an Instrument. In the second track's "MIDI From" menu, select the track containing Skald.

- **Logic Pro (AU)**: Load Skald as a "MIDI FX" (found above the Instrument slot in the mixer).
- **Cubase/Nuendo**: Load Skald on a MIDI track. On your Instrument track, set the "MIDI Input" to "Skald - MIDI Out."
- **FL Studio**: Load Skald in the Wrapper. In the Wrapper settings (cog icon), set the "Output Port" to (e.g.) Port 1. On your target synthesizer, set the "Input Port" to Port 1.
- **Reaper**: Place Skald before an instrument in the FX chain, or use "Sends" to route MIDI to different tracks.
- **Studio One (PreSonus)**: Load Skald on a MIDI/Instrument track. create a second Instrument track for your sound source. On the second track, set the **Instrument Input** to "Skald" (instead of "All Inputs" or your MIDI keyboard). Ensure the Record Arm or Monitor button is active on both tracks.
- **Samplitude / Sequoia (Magix)**: Insert Skald as a VSTi on a track. On your target Instrument track, go to the **MIDI Input** settings in the Track Inspector. Choose **VSTi Out > Skald** as the source. Enable MIDI monitoring (the small speaker icon) to hear the generated notes.

  (These DAWs are geared towards mastering/editing; the standard procedure via "VSTi Out" is how they handle internal MIDI routing.)
- **Pro Tools (Avid)**: Create a **MIDI Track** and insert Skald. Create an **Instrument Track** with your desired plugin (e.g., Xpand!2). In the MIDI Track's **Output selector**, choose the name of your Instrument Track and the specific MIDI channel you wish to trigger. Note: Since Pro Tools handles MIDI FX differently, using Skald as a MIDI generator on its own track is the most stable method.

  (Pro Tools doesn't like plugins that generate MIDI on the same track, so the two-track solution (MIDI + Instrument) is the one that guarantees the least problems for the user).
- **Reason (Reason Studios)**: Create Skald in the Rack. Create your target Instrument (e.g., Europa or Thor). In the Rack, use the **CV/MIDI routing** on the back of the devices or use a "MIDI In Device" to route Skald's output to the instrument's sequencer input.

Make sure the destination tracks do not have "MIDI Thru" enabled to avoid infinite note loops if Skald is set to also receive external inputs (above all in Pro Tools and in Samplitude).

Best Practices: MIDI Channel Organization in Skald

We conclude this guide with a best practices framework for assigning MIDI channels. Following a standard, even if arbitrary, will help you stay on track when your projects become complex (with 20 or 30 dots on disk).

While you're free to do whatever you want, adopting a structure inspired by "General MIDI" or professional workflows will save you hours of troubleshooting.

Canale MIDI	Destinazione Tipica	Tipo di Dot consigliato
Canale 1	Bass / Sub-Bass	Short gate, high velocity.
Canale 2-9	Lead / Keys / Arp	Variable gate, uses Swing a lot.
Canale 10	**Drum Kit (Standard)**	Variable velocity for "Ghosting".
Canale 11-15	Percussioni / FX	Randomize and Low Probability.
Canale 16	Utility / Clock Out	Only Dots placed on the quarters.

💡 Tips for a Clean Workflow

1. **Colore e Canale**: Anche se Skald assegna i colori in base all'anello, cerca di raggruppare visivamente i Dots. Ad esempio, potresti decidere che i primi 3 anelli interni (i più piccoli) siano sempre sul Canale 10 per la batteria.
2. **The HUD** is your friend: When you use **ALT + Drag**, the HUD displays the channel number. If you're driving a multi-timbral sampler (like the MPC or Kontakt), make sure the number in the HUD matches the instrument's "MIDI Slot."
3. **Channel Gate Management:**
   - On the **Synth** channels (**Lead/Pad**), experiment with **CTRL + Drag** to create very long arcs. This will create interesting legato lines as the record spins.
   - On the **Drum channels (Ch 10)**, keep the Gate at a minimum (almost invisible arc) to avoid sample overlap that could "bog down" the sound.
4. **Export**: If you choose to drag the generated MIDI into your DAW, remember that the resulting clip will be either "Type 0" or "Type 1." DAWs will typically automatically separate notes onto different tracks based on channel if you import them correctly.

6. Technical Insights & Logic

Skald is built on the **JUCE Framework** using a high-precision sample-accurate timing engine.

- **The Rotation Engine:** Instead of a standard "grid," Skald calculates triggers based on the floating-point angle of the dots relative to the current playhead position. This allows for the **Swing** and **Reverse** functions to work seamlessly without losing sync.
- **The Hyper-Euclidean Algorithm:** Uses a modified Bjorklund algorithm that doesn't just calculate linear pulses, but maps them onto a polar coordinate system, allowing for "spiral" patterns when the **Depth** parameter is increased.
- **UI Rendering:** Uses Hardware Acceleration (OpenGL/CoreGraphics) to ensure the 60fps rotation of the turntable does not strain the CPU, leaving more power for your synthesizers.
- **Export Logic:** The MIDI Export function translates the polar coordinates (angle and ring index) of every dot into a linear MIDI sequence. It calculates the exact onset and duration based on the current BPM and Speed settings, ensuring that the exported file perfectly matches the "groove" you heard during the live generation.


Options
Skald
Penta
SCALE
C
KEY
1x
DIV
0
OCT
PLAY
TEMPO
120
PROB
SWING
VEL
VVAR
GATE
ALGORITHMIC GEN
TARGET RING
Ring 1
PULSES
4
STEPS
16
HYPER-DEPTH
1
SHIFT
0
GENERATE
REV
MOTOR
REFRESH
ADD
RAND
CLR
SAVE
MID
LOAD
HELP
v1.1.5